

PROFILE

Python platform engineer focused on ML and data-platform reliability, with experience building Dagster, AWS, and containerized systems for operational forecasting, model retraining, lakehouse pipelines, CI/CD, and developer workflows.

SELECTED PLATFORM PROJECT

Energy Market Delta Lake (AWS/Dagster Proof of Concept)

Demo | [GitHub](#)

- Built a public AWS/Dagster lakehouse for Australian energy data using Pulumi, VPC/IAM/ECR/ECS, S3 Delta/Parquet storage, DynamoDB Delta locking, Postgres metadata, Caddy/auth, and Marimo dashboards.
- Implemented 200+ assets and checks across scheduled discovery, event-driven ingestion, bronze Delta tables, source-specific silver snapshots, gas-model marts, sensors, schedules, and maintenance jobs.
- Added Podman/LocalStack local parity, cached archive seeds, Dagster GraphQL e2e launches, dependency-wave asset batches, asset-check monitoring, and regression budgets.

PLATFORM EXPERIENCE

Data Scientist, AEMO

Nov 2025 – Present

NEM Operational Forecasting

- Refactored and operated a Python/Dagster ETL with around 50 assets for high-frequency operational forecasting workloads.
- Stabilized minute-level batch jobs by fixing networking/runtime issues and migrating Dagster metadata from SQLite to Postgres.
- Built Pyinfra, Podman, and multi-repository gRPC deployment patterns to move delivery toward self-hosted Linux CI/CD.
- Strengthened Azure DevOps quality gates with Bash, Ruff, Prek, Pytest, Mypy, and reusable asset factory patterns.
- Retrained and deployed Frequency Uncertainty Models; fed BOM ensemble data into ML forecasting pipelines.

Market Analyst & Analytics Developer, Energy Australia

Jun 2022 – Nov 2023

Trading Data & Systems

- Built a PySpark, Airflow, and Databricks Delta ingestion pipeline for Plexos forecasts, reducing trader analysis time by more than eight hours.
- Added S3-triggered ingestion, New Relic logging and alerts, Terraform, Azure Pipelines, Python package deployment, automated tests, coverage, and SonarQube reporting.
- Implemented hedging position logic in Databricks and scheduled Airflow DAGs across accounts.

Earlier Platform and Analytics Roles

2018 – 2025

AEMO Gas, Origin Energy, University of Melbourne

- Automated AEMO/AER data validation, SQL Server upgrade checks, Redshift integrations, and Marimo QA notebooks; used Pytest to compare UAT and production stored procedure outputs.
- Delivered Salesforce, SAP, IVMS, and telemetry analytics that supported LPG stock reconciliation and about \$500,000 in asset-use savings.
- Developed C++/Python scientific computing code on HPC systems and trialed CNNs for OCT image segmentation and shear stress prediction.

SKILLS

Platform/cloud	AWS, Pulumi, Terraform, ECS/Fargate, ECR, IAM, VPC, S3, DynamoDB, Postgres, LocalStack
MLOps/data	Dagster, Airflow, Databricks, Delta Lake, PySpark, Polars, Pandas, model retraining and deployment
DevEx/quality	Python, uv, Pytest, Ruff, Prek, Mypy, Zuban, Bash, Azure DevOps, CI/CD, GitHub, Codex
Containers	Podman, Docker-compatible workflows, Linux services, Pyinfra, Caddy, FastAPI

EDUCATION

Masters of Engineering (With Distinction), The University of Melbourne	2015 – 2017
Bachelor of Science, The University of Melbourne	2012 – 2014